C28 Amines – Knowledge Organiser

AQA A-level Chemistry • Set 1: Structure and Basicity **Paper 2**

#	Question	Answer
1	Primary amine	N bonded to one carbon
2	Secondary amine	N bonded to two carbons
3	Tertiary amine	N bonded to three carbons
4	Quaternary ammonium ion	N bonded to four carbons, positive charge
5	Why amines are bases	Lone pair on N accepts H^+
6	Why amines are nucleophiles	Lone pair on N
7	Order of base strength	Primary aliphatic amine > NH_3 > aromatic amine
8	Why aliphatic amines are stronger bases	Alkyl groups push electron density onto N
9	Why aromatic amines are weaker bases	Lone pair delocalised into ring
10	Use of quaternary ammonium salts	Cationic surfactants (fabric softeners)

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AQA A-level Chemistry • Set 2: Preparation and Reactions **Paper 2**

#	Question	Answer
1	Amine from a halogenoalkane	Excess NH_3 in ethanol, heat, sealed tube
2	Amine from a nitrile	$LiAlH_4$ or H_2/Ni
3	Why the nitrile route is better	Gives only the primary amine
4	Reagents to reduce nitrobenzene	Sn + conc HCl, then NaOH
5	Equation for reducing nitrobenzene	$C_6H_5NO_2 + 6[H] \rightarrow C_6H_5NH_2 + 2H_2O$
6	Product of reducing nitrobenzene	Phenylamine
7	Use of aromatic amines	Making dyes
8	Amine + acid	Ammonium salt
9	Product of ethylamine + HCl	Ethylammonium chloride
10	Amine + acyl chloride	*N*-substituted amide